CEL: the two operating modes

2026-10-13

Notation follows `problem_setup.md`. Constants are the paper's (`sprt_posterior.py`, pinned by `tests/test_sprt_posterior.`
Each statement is labelled **exact** or **model-based**.

1. Common ground

Cases and target. A test batch $\mathcal{D}_{\text{test}} = \{X_j\}_{j=1}^{m}$ ($m = 50$) of binary questions, a black-box model f, and a selection size K ($K = 10$). For case j, $\hat{Y}_j$ is the majority answer of its draws and $Z_j = \mathbb{1}\{\hat{Y}_j = Y_j\}$ its correctness, the only target.

Ledger. Every call on case j appends a pass/fail outcome to its evidence E_j:

- *draws* on fresh paraphrases, $A_{j1}, A_{j2}, \cdots \in \{0, 1\}$ (agree with $\hat{Y}_j$ or not). With one paraphrase per draw they are exactly i.i.d. Bern$(\bar{p}_j)$, where $\bar{p}_j$ is the case's stability (Lemma 1);
- *probes* $g \in \mathcal{F} = \{\text{negation}, \text{relabel}, \text{distractor}\}$, each pass/fail;
- *other signals*: challenge (1 call), rationale entropy (1 judge call), and verbal confidence and text signals (free when they ride on the draws).

One generator call per case produces the paraphrases.

Score model (model-based). On the labelled fitting split $\mathcal{D}_{\text{fit}}$, replayed under the same policy, a ridge logistic regression of Z on features $x(E)$ gives

$$\hat{\mu}(E) = \sigma(\hat{\beta}^\top x(E)), \qquad s(E) = \hat{\mu}(E) - \hat{\lambda}\,\hat{\sigma}(E),$$

where $\hat{\sigma}$ combines the fit's uncertainty, the stability term's posterior variance, and the variance of unbought signals. $\hat{\lambda} \geq 0$ is chosen by cross-validated prec@K on $\mathcal{D}_{\text{fit}}$; $\hat{\lambda} = 0$, ranking by $\hat{\mu}$, is allowed. $\hat{\mu}$ is an estimate of $P(Z = 1 \mid E)$; s is a ranking score.

Conformal layer (exact; optional in either mode). With the calibration split $\mathcal{D}_{\text{cal}}$ replayed under the same policy,

$$p_j = \frac{1 + \#\{i \in \mathcal{D}_{\text{cal}} : Z_i = 0,\ s_i \geq s_j\}}{n_{\text{cal}} + 1},$$

and Benjamini–Hochberg at level α on $(p_j)_{j \leq m}$ returns a set $\mathcal{S}_\alpha$ with FDR$(\mathcal{S}_\alpha) \leq \alpha$ (Jin & Candès 2023). The guarantee is on correctness and marginal over calibration draws. It requires each case's calls to depend only on that case's own outcomes, so that calibration and test scores are exchangeable. Both modes below satisfy this; the frontier allocation, whose calls depend on the rest of the batch, does not.

2. CEL-k: ranking mode (the budget is fixed)

Input. A budget $\bar{k}$: calls per case, averaged over the batch.

Plans. A plan $a = (k_1, \text{para}, e, F, \text{chal}, \text{se})$ screens every case with k_1 draws. A case whose screen is unanimous (it *survives*) gets e more draws, the probes $F \subseteq \mathcal{F}$, and optionally the challenge and the rationale entropy. Case j costs

$$C_j(a) = \text{para} + k_1 + \mathbb{1}\{j \text{ survives}\}\,(e + |F| + \text{chal} + \text{se}),$$

with expected cost $\bar{C}(a) = \text{para} + k_1 + \pi_{k_1}\,(e + |F| + \text{chal} + \text{se})$, where π_{k_1} is the screen's survival rate.

Plan choice (on dev only). Over the replayable plan space $\mathcal{A}$, with $\widehat{C}_{\text{dev}}$ and $\widehat{\text{prec}}_{\text{dev}}$ measured on the dev slice,

$$\hat{a}(\bar{k}) = \arg \max_{a \in \mathcal{A}:\ \widehat{C}_{\text{dev}}(a) \le \bar{k}} \widehat{\text{prec@}K}_{\text{dev}}(a),$$

with ties going to the cheaper plan. The dev score ranks by Borda over the signals, the label-light criterion the plan space was designed with.

Output. Fit $(\hat{\beta}, \hat{\lambda})$ on $\mathcal{D}_{\text{fit}}$ under $\hat{a}(\bar{k})$, run the plan on the batch, and deliver

$$\mathcal{S}_{\bar{k}} = \text{top-}\,K_j\ s(E_j) \qquad (|\mathcal{S}_{\bar{k}}| = K,\ \text{random tie-breaks}).$$

Properties.

- *Budget*: $\frac{1}{m}\sum_j C_j(\hat{a}) \approx \widehat{C}_{\text{dev}}(\hat{a}) \le \bar{k}$, in expectation, not per case.
- *Optimality* (model-based): if accuracy is monotone in s (Assumption 1), the top K maximise the expected number of correct selections (Prop. 1).
- *Per-answer guarantee*: none. For a set-level guarantee, add the conformal layer.

3. Certified CEL: adaptive mode (the evidence standard is fixed)

Input. An evidence standard $(\varepsilon_0, p_0^*) = (0.05, 0.8)$ and the SPRT $(p_0, p_1, \alpha_s, \beta_s) = (0.7, 0.95, 0.1, 0.1)$.

Certificate (exact). Under the Jeffreys prior $\bar{p} \sim \text{Beta}(\frac{1}{2}, \frac{1}{2})$, after m of n draws agree,

$$\varepsilon(m, n) = P(\bar{p} < p_0^* \mid m \text{ of } n) = I_{p_0^*}(m + \tfrac{1}{2},\ n - m + \tfrac{1}{2}).$$

The certification depth $d^* = \min\{d : \varepsilon(d, d) \le \varepsilon_0\} = 9$ (Cor. 1; $\varepsilon(9, 9) = 0.042$).

Policy, per case (the truncated SPRT):

1. One generator call, then a screen of $s_1 = 3$ draws.
2. If the screen is not unanimous, stop: the case costs 4 calls and is ranked by its screen agreement.
3. Otherwise run stage 2: draws up to the depth d^* (6 more), the probe battery $\mathcal{F}$, the rationale entropy and the challenge. A survivor costs 15 calls (16 with a separate verbal call).

The spend is set by the data, with expectation $\mathbb{E}[C] = 4 + \pi_3 \cdot 11$ (or $\cdot 12$), where $\pi_3 = \mathbb{E}[\bar{p}^3]$. At the paper's survival rates the mean is about 9–13 calls.

Output.

$$\mathcal{S}_{\text{cert}} = \{\, j : j \text{ survives},\ m_j = n_j,\ \varepsilon(m_j, n_j) \le \varepsilon_0 \,\},$$

each delivered answer with its ledger statement. The batch is also ranked: by the cascade key, or by $s(E)$ fitted under this policy. With the conformal layer on that score, it also delivers $\mathcal{S}_\alpha$.

Properties.

- *Per-answer guarantee* (exact, given Lemma 1): for every $j \in \mathcal{S}_{\text{cert}}$, $P(\bar{p}_j < p_0^* \mid E_j) \leq \varepsilon_0$. This is a statement about stability, not correctness: a stable answer can be wrong (Lemma 4). The probes, and the conformal layer on s, are what reach correctness.
- *Model-based variant*: counting each passed probe as κ_g draws (Prop. 3) certifies earlier, but only under that model.
- *Set-level guarantee*: FDR($\mathcal{S}_\alpha$) $\leq \alpha$ (exact), as in §1.

4. How the modes relate

	CEL-k	Certified CEL
Fixed	budget $\bar{k}$	standard (ε_0, p_0^*)
Calls per case	$\leq \bar{k}$ on average	4 or 15–16 (data-dependent)
Delivers	$\mathcal{S}_{\bar{k}}$, exactly K	$\mathcal{S}_{\text{cert}}$, 0 to m
Per-answer guarantee	none	stability certificate (exact)
Set-level FDR (optional)	$\mathcal{S}_\alpha$ on s (exact)	$\mathcal{S}_\alpha$ on s (exact)
Use	cost-capped triage	auto-accept with an evidence record

- **Nesting.** Certified CEL's policy is one plan of CEL-k's space, $a_{\text{CEL}} = (3, \text{para}, 6, \mathcal{F}, 1, 1) \in \mathcal{A}$. CEL-k at $\bar{k} = \bar{C}(a_{\text{CEL}})$ can therefore replay it, and the budget curve meets certified CEL's point there.
- **Certification budget** (Cor. 2). No plan with $\bar{C}(a) < B^*$ can deliver an exact certificate, where B^* is the least expected cost of a plan that reaches d^* agreeing draws. Below B^*, CEL-k never certifies; the conformal layer still applies.
- **Elbow.** Let $(\bar{k}_i, y_i)$ be CEL-k's dev curve over the budget grid, with y_i the running maximum of $\widehat{\text{prec@}K}_{\text{dev}}(\hat{a}(\bar{k}_i))$. With both coordinates scaled to $[0,1]$ ($\tilde{k}$, $\tilde{y}$), the elbow is $k^* = \arg\max_i (\tilde{y}_i - \tilde{k}_i)$ (Kneedle; ties go to the smaller budget). It is chosen on dev only. The metrics table compares both modes there: CEL-k and every baseline at k^* calls, and certified CEL at its own spend.